

graphic formats ranging from neuro-lucida reconstructions, lines, triangles, particles to NURBS. Ultra-high-resolution graphics are designed by deploying interactive walk inside and navigation technologies. 2D, 3D and 3D immersive visualisation systems are also likely to be used.

BlueImage is a software module connected to BlueVision that permits *in silico* imaging of the activity produced in the neocortical column. All values that emerge as an output of the simulation can be imaged. Microcircuit analysis is possible as Blue Gene simulations can generate terabytes of data in a span of minutes of simulations.

BlueAnalysis is used to graphically display, analyse, discard and archive data at a speed as high as feasible. Microcircuit simulations are feasible since the neocortical microcircuit is simulated via NeoCortical Simulator, which is capable of large-scale simulations. NeoCortical Simulator is optimised for parallel simulations with MPI messaging and permits easy expansion of the complexity of the simulation.

NeoCortical Simulator is deployed for all large-scale simulations using the least neuronal models. As many as ten compartmental models can be used in NeoCortical Simulator. Detailed neuron simulations are conducted via NEURON developed and implemented for simulations on Blue Gene. A merged NeoCortical Simulator-NEURON simulator called Neurodamus carries out large-scale NEURON simulations. Neurodamus is likely to evolve further for optimal large-scale Blue Gene simulations of multi-compartmental neuron models.

BlueStim is a software interface to Neurodamus that enables mapping of external input into any one of the 100 million synapses in the



Fig. 8: Neuro-Robotics project aims to understand essential brain mechanisms (Image courtesy: neurorobot.kaist.ac.kr)



Fig. 9: In the future, we might be able to upload our memories and personalities to virtual avatars, after our death (Image courtesy: www.journal.com.ph)

column. Stimulus Generator permits connecting of columns with the external world or with other columns and other brain regions.

BlueRead, a software interface to NeoCortical Simulator and Neurodamus, enables us to define values that are to be read out of the simulator for visualisation, display and analysis.

What the future holds

The brain carries out several analogue operations that computers are not capable of performing and, in many cases, it manages to carry out hybrid digital-analogue computing. The biggest differentiating factor between the brain and computers is that the brain is constantly changing with time. Imagine, if components such as integrated circuits and transistors started changing, then a computer would actually end up becoming unusable. You can imagine the brain as a dynamically-morphing computer, considerably different from other organs like the

heart or lungs.

Understanding the brain is vital, not just to understand the biological mechanisms that give us our thoughts and emotions, and that make us human, but for practical reasons. Understanding how it processes information can make a fundamental contribution to the development of new computing technologies—neuroinformatics and neuromorphic computing. More important still, understanding the brain is essential to understand, diagnose and treat brain diseases that are imposing a rapidly-increasing burden on the world's aging population.

The present configuration of Blue Gene, that is, BlueGene/Q, is capable of touching pet flops speeds and quadrillions of calculations

per second. This supercomputer has as many as 65,536 cores and extended memory capabilities. The next generation of Blue Gene supercomputers is expected to be able to deliver an even higher level of computing power and, hence, be able to simulate even more neurons with significant complexity.

However, sky is the limit, as today scientists are thinking beyond all this and conducting challenging research studies. They are hoping to create an artificial brain that can think, respond, take decisions and store any type of data in its memory. Key objective is to upload a human brain into a machine, which will enable man to think and take decisions effortlessly.

After the death of the human body, the virtual brain will actually be able to act as the deceased. Hence, there will be no loss of knowledge, intelligence, personalities, feelings and memories of humans. In a way, man is on his way to becoming immortal. **EFY**